

Problem 1 (§10.4#14). Let I be a nonempty index set and, for each $i \in I$, let N_i be a left R -module. Let M be a right R -module. Prove the group isomorphism $M \otimes (\bigoplus_{i \in I} N_i) \cong \bigoplus_{i \in I} (M \otimes N_i)$.

Solution. Define the map $\phi: M \times (\bigoplus_{i \in I} N_i) \rightarrow \bigoplus_{i \in I} (M \otimes N_i)$ by

$$\phi(m, (n_i)_{i \in I}) = (m \otimes n_i)_{i \in I}.$$

This map is clearly additive in each argument and

$$\phi(mr, (n_i)_{i \in I}) = (mr \otimes n_i)_{i \in I} = (m \otimes rn_i)_{i \in I} = \phi(m, r(n_i)_{i \in I}),$$

so ϕ is R -balanced, thus inducing a homomorphism $\tilde{\phi}: M \otimes (\bigoplus_{i \in I} N_i) \rightarrow \bigoplus_{i \in I} (M \otimes N_i)$ which is uniquely determined by $\tilde{\phi}(m \otimes (n_i)_{i \in I}) = (m \otimes n_i)_{i \in I}$.

In the other direction, for each $\alpha \in I$, the map $\psi_\alpha: M \times N_\alpha \rightarrow M \otimes (\bigoplus_{i \in I} N_i)$ given by $\psi_\alpha(m, n) = m \otimes \iota_\alpha(n)$, where $\iota_\alpha: N_\alpha \rightarrow \bigoplus_{i \in I} N_i$ is the canonical embedding of N_α into the direct sum, is R -balanced. This means ψ_α induces a homomorphism $\tilde{\psi}_\alpha: M \otimes N_\alpha \rightarrow M \otimes (\bigoplus_{i \in I} N_i)$ which is determined by $m \otimes n \mapsto m \otimes \iota_\alpha(n)$. Then, the $\tilde{\psi}_\alpha$ canonically induce a map $\tilde{\psi}: \bigoplus_{i \in I} (M \otimes N_i) \rightarrow M \otimes (\bigoplus_{i \in I} N_i)$ determined by

$$\tilde{\psi}((m_i \otimes n_i)_{i \in I}) = \sum_{i \in I} m_i \otimes \iota_i(n_i)$$

where $n_i \in N_i$ and cofinitely many of the $m_i \otimes n_i$, hence cofinitely many of the $m_i \otimes \iota_i(n_i)$ summands, are zero.

We show that $\tilde{\phi}$ and $\tilde{\psi}$ are inverses. For sequences of simple tensors $(m_i \otimes n_i)_{i \in I}$ in $\bigoplus_{i \in I} (M \otimes N_i)$ with $m_i \otimes n_i = 0$ for cofinitely many $i \in I$,

$$\tilde{\phi} \circ \tilde{\psi}((m_i \otimes n_i)_{i \in I}) = \sum_{i \in I} \tilde{\phi}(m_i \otimes \iota_i(n_i)) = \sum_{i \in I} m_i \otimes n_i = (m_i \otimes n_i)_{i \in I}$$

(To be precise, the summands in the latter sum are really elements of the embedding of $M \otimes N_i$ into the direct sum of tensors.) A generic element of the direct sum of $M \otimes N_i$ is given by a finite sum of simple tensors at finitely many indices $i \in I$, which can be written as the sum of elements of the form $(m_i \otimes n_i)_{i \in I}$ as in the equation above, so $\tilde{\phi} \circ \tilde{\psi}$ is the identity on $\bigoplus_{i \in I} (M \otimes N_i)$. Conversely, for simple tensors $m \otimes (n_i)_{i \in I} \in M \otimes (\bigoplus_{i \in I} N_i)$,

$$\tilde{\psi} \circ \tilde{\phi}(m \otimes (n_i)_{i \in I}) = \tilde{\psi}((m \otimes n_i)_{i \in I}) = \sum_{i \in I} m \otimes \iota_i(n_i) = m \otimes \left(\sum_{i \in I} \iota_i(n_i) \right) = m \otimes (n_i)_{i \in I},$$

implying that $\tilde{\psi} \circ \tilde{\phi}$ is the identity on $M \otimes (\bigoplus_{i \in I} N_i)$, and proving that $\tilde{\phi}$ and $\tilde{\psi}$ are isomorphisms. $\square$

Problem 2 (§10.4#16). Suppose R is commutative and let I and J be ideals of R , so R/I and R/J are naturally R -modules.

- (a) Prove that every element of $R/I \otimes_R R/J$ can be written as a simple tensor of the form $(1+I) \otimes (r+J)$.
- (b) Prove that there is an R -module isomorphism $R/I \otimes_R R/J \cong R/(I+J)$ mapping $(r+I) \otimes (r'+J)$ to $rr' + (I+J)$.

Solution. (a) Any simple tensor can be written as $(a+I) \otimes (b+J) = (1+I) \otimes (ab+J)$, so a generic element of $R/I \otimes_R R/J$, which is a sum of finitely many simple tensors, can be written as

$$\sum_{i=1}^m (1+I) \otimes (r_i+J) = (1+I) \otimes \left(\sum_{i=1}^m r_i + J \right).$$

(b) It is easy to see that the map $(r+I, r'+J) \mapsto rr' + (I+J)$ from $R/I \times R/J$ to $R/(I+J)$ is R -balanced (since $(sr+I, r'+J)$ and $(r+I, sr'+J)$ both map to $sr r' + (I+J)$), so the given map, call it ϕ , is a well-defined homomorphism of R -modules.

We will show that the map $\psi: R/(I+J) \rightarrow R/I \otimes_R R/J$ defined by $\psi(r+(I+J)) = (1+I) \otimes (r+J)$ is the inverse of ϕ , so that ϕ is indeed an isomorphism of R -modules. First, to see that ψ is well-defined, we note that the map $R \rightarrow R/I \otimes_R R/J$ defined by $r \mapsto (1+I) \otimes (r+J)$ has both I and J , hence $I+J$, as a subset of its kernel, since $r \in I$ implies $(1+I) \otimes (r+J) = (r+I) \otimes (1+J) = 0_{R/I} \otimes (1+J) = 0$ and $r \in J$ implies $r+J = 0_{R/J}$, so this map on R factors through the quotient $R/(I+J)$ to yield ψ . Now we check that ϕ and ψ are inverses:

$$\begin{aligned} \psi \circ \phi((1+I) \otimes (r+J)) &= \psi(r+(I+J)) = (1+I) \otimes (r+J), \\ \phi \circ \psi(r+(I+J)) &= \phi((1+I) \otimes (r+J)) = r + (I+J) \end{aligned}$$

so $\psi \circ \phi$ and $\phi \circ \psi$ are respectively the identities on $R/I \otimes_R R/J$ and $R/(I+J)$, where we use part (a) to ensure that we only need to check that simple tensors of the form $(1+I) \otimes (r+J)$ get sent to themselves by $\psi \circ \phi$. Thus, ϕ is indeed an isomorphism of the given R -modules. $\square$

Problem 3 (§10.4#21). Suppose R is commutative and let I and J be ideals of R .

- (a) Show that there is a surjective R -module homomorphism from $I \otimes_R J$ to the product ideal IJ mapping $i \otimes j$ to the element ij .
- (b) Give an example to show that the map in (a) need not be injective.

Solution. (a) The map $I \times J \rightarrow IJ$ given by $(i, j) \mapsto ij$ is clearly additive in both arguments and satisfies $(ir, j) = (i, rj)$, so it is R -balanced; thus, the map $\phi: I \otimes_R J \rightarrow IJ$ given by $\phi(i \otimes j) = ij$ is a well-defined homomorphism of R -modules. Recall that IJ is generated as an R -module by finite sums of the form $\sum_{k=1}^m i_k j_k$ for $i_k \in I, j_k \in J$ for $k = 1, \dots, m$, so it suffices to show that all such finite sums are in the image of ϕ . Indeed, for a given sum $\sum_{k=1}^m i_k j_k$,

$$\phi \left(\sum_{k=1}^m i_k \otimes j_k \right) = \sum_{k=1}^m i_k j_k,$$

so $\phi(I \otimes_R J) = IJ$.

(b) Exercise 17 in this section shows that, with $R = \mathbb{Z}[x]$ and $I = J = (2, x)$, the elements $2 \otimes x, x \otimes 2 \in I \otimes_R I$ are distinct, while $\phi(2 \otimes x) = \phi(x \otimes 2) = 2x$. Thus, ϕ has nontrivial kernel, for example containing the nonzero element $2 \otimes x - x \otimes 2$. $\square$

Problem 4 (§10.5#2). Suppose that

$$\begin{array}{ccccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C & \xrightarrow{h} & D \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \xrightarrow{h'} & D' \end{array}$$

is a commutative diagram of groups and that the rows are exact. Prove that

- (a) if α is surjective, and β, δ are injective, then γ is injective.
- (b) if δ is injective, and α, γ are surjective, then β is surjective.

Solution. (a) Suppose $c \in \ker \gamma$; we will show $c = 1$. Then $\delta h(c) = 1$ and δ is injective, so $h(c) = 1$, meaning $c \in \ker h = \text{im } g$. Write $c = g(b)$ for $b \in B$; then

$$1 = \gamma(c) = \gamma g(b) = g' \beta(b),$$

so $\beta(b) \in \ker g' = \text{im } f'$. Write $\beta(b) = f'(a')$ for $a' \in A'$; since α is surjective, $a' = \alpha(a)$ for some $a \in A$. Thus $\beta(b) = f' \alpha(a) = \beta f(a)$; by the injectivity of β , this means $f(a) = b$, i.e. $b \in \text{im } f = \ker g$. Therefore $c = g(b) = 1$, i.e. $\ker \gamma$ consists of only the identity.

(b) Suppose $b' \in B'$. By surjectivity, $g'(b') = \gamma(c)$ for $c \in C$. By exactness at C' , $\gamma(c) = g'(b') \in \ker h'$, hence

$$1 = h' \gamma(c) = \delta h(c),$$

implying by injectivity that $h(c) = 1$. By exactness at C , $c = g(b)$ for some $b \in B$. Observe that

$$g'(\beta(b) \cdot (b')^{-1}) = \gamma g(b) \cdot g'(b')^{-1} = \gamma(c) \cdot \gamma(c)^{-1} = 1,$$

so $\beta(b) \cdot (b')^{-1} \in \ker g' = \text{im } f'$, hence equals $f'(a')$ for some $a' \in A'$; surjectivity of α then implies that $a' = \alpha(a)$ for some $a \in A$. That is, $\beta(b) \cdot (b')^{-1} = f' \alpha(a) = \beta f(a)$ so $\beta(b \cdot f(a)^{-1}) = b'$, meaning b' is in the image of β . $\square$

Problem 5 (§10.5#5). Prove that an arbitrary direct sum $\bigoplus_{i \in I} A_i$ of R -modules is flat if and only if each A_i is flat.

Solution. To show flatness of an R -module A , it suffices to show that for an injective R -module homomorphism $f: L \rightarrow M$, $f \otimes 1: L \otimes_R A \rightarrow M \otimes_R A$ is also injective, so fix an injective map $f: L \rightarrow M$.

If $\bigoplus_{i \in I} A_i$ is flat, for a given R -module B and index $i \in I$ let $\iota_{B,i}: B \otimes_R A_i \rightarrow B \otimes_R (\bigoplus_{i \in I} A_i)$ be the canonical embedding of $B \otimes_R A_i$ into $\bigoplus_{i \in I} (B \otimes_R A_i)$, composed with the isomorphism $\phi_B: \bigoplus_{i \in I} (B \otimes_R A_i) \rightarrow B \otimes_R (\bigoplus_{i \in I} A_i)$ defined in the first problem. Note that $\iota_{B,i}$ is injective for each B and that we have a commutative diagram

$$\begin{array}{ccc} L \otimes_R A_i & \xrightarrow{f \otimes 1} & M \otimes_R A_i \\ \downarrow \iota_{L,i} & & \downarrow \iota_{M,i} \\ L \otimes_R (\bigoplus_{i \in I} A_i) & \xrightarrow{f \otimes 1} & M \otimes_R (\bigoplus_{i \in I} A_i). \end{array}$$

Since the morphism $(f \otimes 1) \circ \iota_{L,i}$ is equal to $\iota_{M,i} \circ (f \otimes 1)$ and the former, being a composition of injective morphisms, is injective, the latter morphism must also be injective, implying that the upper $f \otimes 1$ is injective, showing that $- \otimes_R A_i$ is left-exact for each $i \in I$.

Conversely, if A_i is flat for each $i \in I$, then $f \otimes 1_i: L \otimes_R A_i \rightarrow M \otimes_R A_i$ is injective for each i where 1_i is the identity on A_i . The injective maps induce injective maps between the coproducts, which, upon passing through ϕ_L and ϕ_M , shows that $f \otimes 1: L \otimes_R (\bigoplus_{i \in I} A_i) \rightarrow M \otimes_R (\bigoplus_{i \in I} A_i)$ is injective. $\square$

Problem 6. This exercise proves Theorem 38 that every left R -module is contained in an injective left R -module.

- (a) Show that M is contained in an injective $\mathbb{Z}$ -module Q .
- (b) Show that $\text{Hom}_R(R, M) \subseteq \text{Hom}_{\mathbb{Z}}(R, M) \subseteq \text{Hom}_{\mathbb{Z}}(R, Q)$.
- (c) Use the R -module isomorphism $M \cong \text{Hom}_R(R, M)$ and the previous exercise to conclude that M is contained in an injective R -module.

Solution. (a) This follows immediately from Corollary 37, which says that the problem holds when $R = \mathbb{Z}$, because all left R -modules are also $\mathbb{Z}$ -modules.

(b) The first inclusion holds because an R -module homomorphism $\phi: R \rightarrow M$ is also a homomorphism of abelian groups if we just ignore the multiplicative structure of the map.

The second inclusion holds because we have an injective map sending morphisms $f \in \text{Hom}(R, M)$ to $\iota \circ f \in \text{Hom}(R, Q)$ where $\iota: M \hookrightarrow Q$ is the inclusion of M into Q .

(c) The preceding exercise shows that $\text{Hom}_{\mathbb{Z}}(R, Q)$ is an injective R -module if Q is, so the chain of inclusions $R \cong \text{Hom}_R(R, M) \hookrightarrow \text{Hom}_{\mathbb{Z}}(R, M) \hookrightarrow \text{Hom}_{\mathbb{Z}}(R, Q)$ from part (b) implies that R is isomorphic to a submodule of the injective left R -module $\text{Hom}_{\mathbb{Z}}(R, Q)$ as desired. $\square$